

(a) Accumulator A_j before the batchcolumn k = position

	1	2	3	4
$p=1$: A	1	0	0	0
$p=2$: B	0	1	0	0
$p=3$: C	0	0	1	0
$p=4$: D	0	0	0	1

row p : item at position p **(b) Votes from one batch**a vote of weight a adds $|a|$ to the column of each item's target positionvote 1 ($a_1 = +1$)

C	A	D	B
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 $\Rightarrow +1$ at (A, 2), (B, 4), (C, 1), (D, 3)vote 2 ($a_2 = -0.5$)

D	B	C	A
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 $a_2 < 0$: the target is reversed

+

reversed ($a_2 < 0$)

A	C	B	D
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 $\Rightarrow +0.5$ at (A, 1), (C, 2), (B, 3), (D, 4)**(c) After the batch**

	1	2	3	4
$p=1$: A	1.5	1	0	0
$p=2$: B	0	1	0.5	1
$p=3$: C	1	0.5	1	0
$p=4$: D	0	0	1	1.5

$$s_p = \frac{\sum_k k A[p, k]}{\sum_k A[p, k]}$$

(d) Row means and reorder

$$s_1 = 1.4 \quad (\text{A})$$

$$s_2 = 3.0 \quad (\text{B})$$

$$s_3 = 2.0 \quad (\text{C})$$

$$s_4 = 3.6 \quad (\text{D})$$

sort by s_p , ties by item ID

$$r_j \leftarrow [A, C, B, D]$$

reset

(e) Reset to identity in the new order

	1	2	3	4
$p=1$: A	1	0	0	0
$p=2$: C	0	1	0	0
$p=3$: B	0	0	1	0
$p=4$: D	0	0	0	1

next batch: $r_j = [A, C, B, D]$